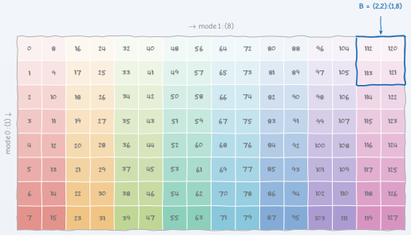
CUTE · LAYOUT · TENSOR · CORE · FLASH · ATTENTION

LEETCUDA

CUDA KERNEL 优化之路

从编程模型到 CUTE 与 FLASHATTENTION



郑 重 声 明
本 书 仅 用 于 开 源 学 习 与 技 术 交 流。
未 经 作 者 本 人 同 意， 不 得 用 于 任 何 商 业 用 途。

LEETCUDA PROJECT (OWN BY DEFRUTII) · 2026 年 4 月

FA2 / FA3 ATTENTION VS CUDNN SDPA

NVIDIA RTX PRO 5000 (SM_120A) - B=1, H=32, N=16384, D=128

CONFIGURATION	TFLOPS	SPEEDUP	RELATIVE
CUDNN SDPA BASELINE	232.4	1.00X	
FA2 MMA			
SK=2, F16ACC	157.9	0.68X	
SK=1, F32ACC	145.6	0.63X	
SK=2, F32ACC	166.5	0.72X	
FA2 CUTE MMA			
SK=1, F32ACC	189.6	0.82X	
SK=2, F32ACC	197.0	0.85X	
FA2 TMA WS (1 WG)			
SK=1, SV=1, F16ACC	161.2	0.69X	
SK=2, SV=1, F16ACC	189.4	0.81X	
SK=2, SV=2, F16ACC	190.5	0.82X	
SK=2, SV=1, F32ACC	204.9	0.88X	
FA3 TMA WS (2 WG)			
SK=1, SV=1, F16ACC	210.1	0.90X	
SK=1, SV=1, F32ACC	210.8	0.91X	
FA2 CUTE TMA WS (1 WG)			
SK=2, SV=1, F32ACC	220.0	0.95X	
SK=3, SV=1, F32ACC	223.4	0.96X	
FA2 CUTE TMA WS PERSISTENT-CTA (1 WG)			
D=128	242.5	1.04X	

HGEMM CUTE SWIZZLE VS CUBLAS

NVIDIA RTX PRO 5000 (SM_120A) - M=N=K=4096, F16ACC; FP8 4096/8192

CONFIGURATION	TFLOPS	SPEEDUP	RELATIVE
CUBLAS BASELINE	236.9	1.00X	
HGEMM CUTE SWIZZLE (F16ACC)			
S=2, BLK_SW=0	229.5	0.97X	
S=2, BLK_SW=1	231.1	0.98X	
S=3, BLK_SW=0	245.6	1.04X	
S=3, BLK_SW=1	246.7	1.04X	
FP8 CUTE GEMM (F32ACC)			
WS S2 128X256, 4096^3	443.7	1.87X	
WS S2 128X256, 8192^3	487.8	2.06X	

SPEEDUP IS RELATIVE TO THE VENDOR BASELINE
(CUBLAS FOR HGEMM, CUDNN SDPA FOR ATTENTION).

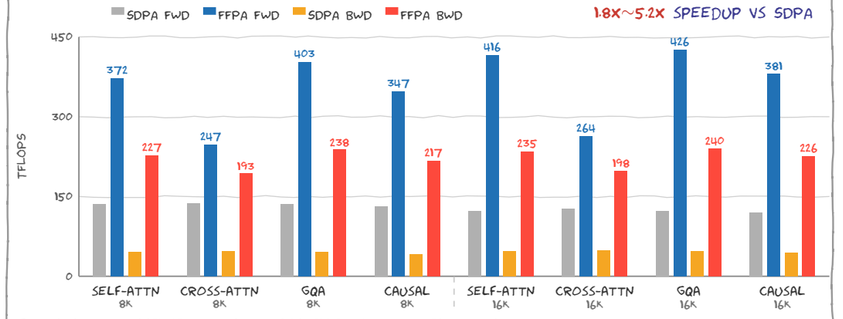
FA SPLIT-D VS CUDNN SDPA

NVIDIA RTX PRO 5000 (SM_120A) - B=1, H=32, N=8192, D=320

CONFIGURATION	TFLOPS	SPEEDUP	RELATIVE
CUDNN SDPA BASELINE	69.8	1.00X	
SPLIT-D TMA WS			
SK=1, SV=1	86.6	1.24X	
SK=2, SV=2	139.9	2.01X	
SPLIT-D TMA NON-WS			
SK=2, SV=2	187.9	2.69X	
SK=2, SV=3	188.8	2.71X	
SK=3, SV=2	204.3	2.93X	

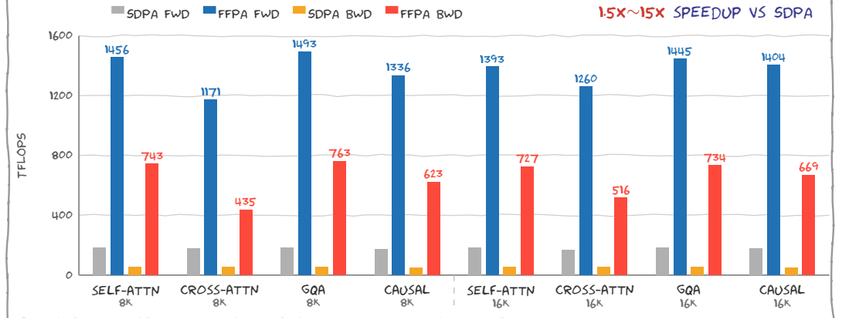
SK = K STAGES (FA), SV = V STAGES (FA), S = STAGES (HGEMM)

FFPA CUTEDSL VS SDPA - H200 (D=512)



LIGHT BARS = SDPA BASELINE; FFPA BAR TOPS LABELLED IN TFLOPS. PEAK 5.2X VS SDPA.

FFPA CUTEDSL VS SDPA - B200 (D=512)



LIGHT BARS = SDPA BASELINE; FFPA BAR TOPS LABELLED IN TFLOPS. PEAK 14.7X VS SDPA.